Bell Work Answer Key

Thursday, October 29 · Module 2, Topic 2, Lesson 3, Day 2 · Writing Equations of Perpendicular Lines

1. Line j has slope $^{2}/_{3}$. Line k is perpendicular to it. What is the slope of line k?

A. 2/3 **B.** 3/2 **C. −3/2 ✓** **D.** −2/3

Negative reciprocal: flip 2/3 → 3/2, change the sign → −3/2. Check: 2/3 · (−3/2) = −1. SE p. 309.

2. What is the slope of any horizontal line?

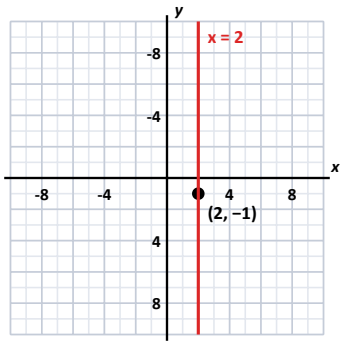
A. undefined **B.** 1 **C.** −1 **D. 0 ✓**

On a horizontal line y never changes, so the rise is 0: slope = 0 ÷ run = 0. (HOY) SE p. 310.

3. Which is the equation of the vertical line through (2, −1)?

A. y = −1 **B. x = 2 ✓** **C.** y = 2 **D.** x = −1

Every point on a vertical line has the same x-value, here 2. (VUX) SE p. 311.



4. Which equation represents a line perpendicular to y = 2x + 3 that passes through the point (4, 1)?

F. y = 2x − 7 **G.** y = ½x − 1 **H.** y = −½x + 1 **J. y = −½x + 3 ✓**

Perpendicular slope −½. Put in (4, 1): 1 = −½(4) + b → 1 = −2 + b → b = 3. y = −½x + 3. SE p. 312.

Go over: Q4 is exactly today's Activity 3.3: find the perpendicular slope, then use the point to find b (or use point-slope: y − 1 = −½(x − 4)). Most common wrong answer: **H** (used the right slope but kept b = 1 from the point).